

# An Euler product for the energy cascade

pole fronts, temperedness, and the limits of the  $L$ -function analogy

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July 2026

## Abstract

To a solution  $u$  of the three-dimensional incompressible Navier–Stokes equations we associate the spectral determinant  $D_u(s, t) = \prod_{j \geq 1} \det(I - A_j(t) \lambda_j^{-s})$ , an Euler-type product over dyadic scales  $\lambda_j = 2^j$  built from finite-rank Littlewood–Paley energy-transfer operators  $A_j(t)$ , together with the meromorphic function  $\Lambda_u = \Gamma_\infty / D_u$ , in loose analogy with completed  $L$ -functions. We prove that for every Leray–Hopf solution on the torus the product converges on a right half-plane, and that along any strong solution the Gevrey regularization of Foias and Temam makes  $D_u(\cdot, t)$  *entire*, so that  $\Lambda_u(\cdot, t)$  is meromorphic on all of  $\mathbf{C}$ : the analytic continuation is a consequence of parabolic smoothing, not of any functional equation. Indeed we show that no functional equation  $\Lambda_u(s, t) = \varepsilon(t) \Lambda_u(1-s, t)$  can hold in the natural Ramanujan-type (“tempered”) normalization of the transfer spectra — and that, whenever some transfer operator has a nonzero eigenvalue, the bare determinant itself admits none, so the failure is independent of the (conventional) archimedean normalization — locating a precise breaking point of the number-theoretic analogy. The temperedness hypothesis that confines the poles is identified exactly: it is a  $\dot{B}_{\infty, \infty}^1$  bound on the velocity, which implies global regularity by a logarithmic Beale–Kato–Majda argument and, conversely, is implied by it — so the determinant repackages, and cannot by itself reduce, the regularity problem, consistent with the blow-up barrier of Tao for averaged Navier–Stokes systems. In the viscous Sabra shell model we compute the transfer matrices in closed form; the per-shell Jacobian block vanishes identically, forcing a windowed construction, with which  $\Lambda_u(\cdot, t)$  is meromorphic on  $\mathbf{C}$  for all  $t$  unconditionally, by the regularity theorem of Constantin, Levant and Titi. Finally we compute the pole structure numerically in a verified statistically steady turbulent state: the pole front advances with abscissa slope  $0.549 \pm 0.007$  in  $\log k_n$ , matching within 1.5% the independently measured typical-amplitude cascade exponent 0.541, so the determinant reads off the intermittency-corrected Kolmogorov exponent; and the spectral lower bound  $\rho(A_n) \geq 0.53 k_n |u_n|$  holds over all 11,000 inertial-range samples, showing that the non-normality of the transfer matrices is bounded. Whether the analogous lower bound holds for the true equations is posed as the main open question: a proof would make pole confinement *equivalent* to regularity, certifying  $\Lambda_u$  as a faithful spectral diagnostic of blow-up — though not, for structural reasons we explain, a mechanism for excluding it.

# 1 Introduction

The three-dimensional incompressible Navier–Stokes equations on the torus  $\mathbf{T}^3 = (\mathbf{R}/2\pi\mathbf{Z})^3$  read

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = u_0, \quad (1)$$

with  $\nu > 0$  and  $u_0$  smooth, divergence-free and mean-free. Whether smooth solutions exist globally in time is the Clay Millennium Problem [5].

The turbulent energy cascade organizes itself across dyadic scales, and it is tempting to compare the resulting hierarchy with the multiplicative hierarchy of primes: local factors at each scale, an Euler-type product, a completed object with (one might hope) a functional equation and a critical strip, and a Ramanujan-type normalization of local spectra. Determinantal generating functions attached to dynamics have a distinguished history — Ruelle’s dynamical zeta functions [11] on one side, Tate’s adelic treatment of Hecke  $L$ -functions [13] on the other — and versions of the analogy for fluid equations circulate informally. The purpose of this paper is to develop the analogy rigorously for (1) and for the Sabra shell model, and, just as importantly, to *delimit* it: to prove exactly which of its promised features hold, for what reason, and which provably fail. We aim for statements that are correct and complete rather than optimistic; in particular, nothing in this paper bears on the Millennium Problem itself, and Section 9 explains structural reasons why no argument of the present type could.

Our contributions are the following.

- (i) **A well-posed construction (§3).** On  $\mathbf{T}^3$  each Littlewood–Paley shell contains finitely many Fourier modes, so the scale-local energy-transfer operators  $A_j(t)$  are finite-rank and the local factors  $\det(I - A_j(t)\lambda_j^{-s})$  are classical determinants, entire in  $s$ ; no zeta-regularization is needed. For every Leray–Hopf weak solution the product  $D_u(s, t) = \prod_{j \geq 1} \det(I - A_j(t)\lambda_j^{-s})$  converges locally uniformly on  $\{\operatorname{Re} s > 11/2\}$ , and all poles of  $\Lambda_u = \Gamma_\infty/D_u$  have abscissae accumulating at most at  $5/2$  (Proposition 3.5). The product starts at  $j = 1$  for a reason we make explicit (Remark 3.3):  $\lambda_0 = 1$  renders the shell-0 factor constant in  $s$ , and retaining it would allow  $D_u(\cdot, t)$  to vanish identically at isolated times.
- (ii) **Meromorphic continuation from smoothing (§4).** Along any strong solution,  $D_u(\cdot, t)$  is *entire* and  $\Lambda_u(\cdot, t)$  is meromorphic on  $\mathbf{C}$  (Theorem 4.1). The mechanism is the instantaneous Gevrey (spatial-analytic) smoothing of Foias–Temam [6]: exponential decay of shell amplitudes forces superexponential decay of  $\|A_j(t)\|$ . Consequently the continuation encodes analyticity of the flow and nothing more (Remark 4.2).
- (iii) **An exact regularity dictionary (§5).** The Ramanujan-type normalization of the transfer spectra — poles confined near  $\operatorname{Re} s = 0$  — is implied by the bound  $\int_0^T \|u(t)\|_{\dot{B}_{\infty,\infty}^1} dt < \infty$ . This bound implies smoothness up to time  $T$ , by a self-contained logarithmic Beale–Kato–Majda argument in the spirit of [2, 8] (Theorem 5.3); and conversely smoothness implies the bound (Proposition 5.4). The hypothesis is therefore *equivalent* to the conclusion: the determinant converts the regularity problem into an equally hard statement about shell amplitudes (Corollary 5.5).
- (iv) **No functional equation (§6).** In the tempered regime, no identity  $\Lambda_u(s, t) = \varepsilon(t)\Lambda_u(1-s, t)$  with  $|\varepsilon(t)| = 1$  can hold; moreover, whenever some transfer operator has a nonzero

eigenvalue, the bare determinant  $D_u$  itself admits no functional equation, so the failure does not depend on the (conventional) archimedean factor (Proposition 6.1, Remark 6.2). This is a precise breaking point of the analogy with Hecke theory: the cascade possesses no self-duality playing the role of Poisson summation, dissipation is not invariant under  $\lambda \mapsto 1/\lambda$ , and energy is additive rather than multiplicative across scales.

- (v) **Explicit transfer matrices for the Sabra model (§7).** We compute the realified Jacobian of the Sabra nonlinearity in closed form (Lemma 7.1). Its per-shell diagonal block *vanishes identically* — the within-shell transfer operator, which is nontrivial for (1) because a Littlewood–Paley shell contains many mutually advecting modes, is zero in any shell model whose triads couple only distinct shells — so the minimal meaningful local factors are windowed compressions  $A_n^{(r)}(t)$  (Definition 7.3). With these, the global regularity and Gevrey estimates of Constantin–Levant–Titi [4] make  $D_u(\cdot, t)$  entire and  $\Lambda_u(\cdot, t)$  meromorphic on  $\mathbf{C}$  for every  $t > 0$ , unconditionally (Theorem 7.5).
- (vi) **The pole front of shell-model turbulence (§8).** In a verified, statistically steady turbulent state (22 shells,  $10^3$  decorrelated snapshots; energy balance closes to 0.6%; the closed-form Jacobian matches finite differences to  $6 \times 10^{-13}$ ), the mean-log spectral radius grows as  $\rho(A_n^{(1)}) \sim k_n^{0.549 \pm 0.007}$  across the inertial range, in agreement within 1.5% with the independently measured typical-amplitude exponent  $d\langle \ln k_n |u_n| \rangle / d \ln k_n = 0.541$ : *the pole front of  $\Lambda_u$  measures the cascade scaling exponent*, including its intermittency correction (the non-intermittent Kolmogorov value is  $2/3$ ). Moreover  $\rho(A_n)/\|A_n\| \geq 0.25$  and  $\rho(A_n) \geq 0.53 k_n |u_n|$  over every inertial-range sample: the non-normality of the transfer matrices is bounded and their spectra are not blind to amplitudes. This answers affirmatively, in the shell-model setting, the spectral lower-bound question (Question 7.7) whose Navier–Stokes form we leave open.

The picture that emerges is double-edged, and we state it plainly. The determinant is a well-defined, computable object whose analytic features are theorems; its pole front carries real physical information (the intermittent cascade exponent); and, granted the open lower bound, pole confinement would be *equivalent* to regularity, making  $\Lambda_u$  a faithful spectral diagnostic of blow-up. What the determinant cannot do is decide regularity: the hypothesis that confines its poles is subcritical by two derivatives relative to the natural scaling of (1), and every estimate we use survives the averaging of Tao [12], whose averaged Navier–Stokes system blows up. Faithful diagnostics are symmetric between regularity and blow-up (§9).

## 2 Preliminaries

### 2.1 Function spaces and Littlewood–Paley decomposition

We work on  $\mathbf{T}^3$  with mean-free vector fields;  $\hat{f}(k)$ ,  $k \in \mathbf{Z}^3$ , denotes Fourier coefficients. The Leray projector onto divergence-free fields is

$$(\widehat{\mathbf{P}v})(k) = \hat{v}(k) - \frac{(k \cdot \hat{v}(k)) k}{|k|^2}, \quad k \neq 0, \quad (2)$$

i.e.  $\mathbf{P}_{ij} = \delta_{ij} - \partial_i \partial_j \Delta^{-1}$ .

Fix a smooth radial  $\varphi$  supported in the annulus  $\{\frac{1}{2} \leq |\xi| \leq 2\}$  with  $\sum_{j \in \mathbf{Z}} \varphi(2^{-j}\xi) = 1$  for  $\xi \neq 0$ , and set  $\Delta_j f = \sum_k \varphi(2^{-j}k) \hat{f}(k) e^{ik \cdot x}$  for  $j \geq 0$  (mean-free fields on the torus need only  $j \geq 0$  after absorbing  $|k| \leq 1$  into  $\Delta_0$ ; we do so silently). Write  $\lambda_j = 2^j$  and  $\tilde{\Delta}_j = \Delta_{j-1} + \Delta_j + \Delta_{j+1}$ . We use throughout the standard Bernstein inequalities on shells (see [1]): for  $f$  with Fourier support in  $\{|k| \sim 2^j\}$ ,

$$\|\nabla f\|_{L^p} \leq C 2^j \|f\|_{L^p}, \quad \|f\|_{L^\infty} \leq C 2^{3j/2} \|f\|_{L^2}, \quad (3)$$

and, for divergence-free  $u$  with vorticity  $\omega = \nabla \times u$ , the two-sided shell estimates

$$c 2^j \|\Delta_j u\|_{L^\infty} \leq \|\Delta_j \omega\|_{L^\infty} \leq C 2^j \|\Delta_j u\|_{L^\infty}, \quad (4)$$

valid because on the annulus  $|k| \sim 2^j$  the curl and its inverse (on divergence-free fields) are Fourier multipliers with symbols smooth and homogeneous of degrees 1 and  $-1$  respectively. Homogeneous Besov norms are  $\|f\|_{\dot{B}_{\infty,\infty}^s} = \sup_j 2^{js} \|\Delta_j f\|_{L^\infty}$ .

## 2.2 Solutions and classical criteria

A *Leray–Hopf weak solution* on  $[0, T)$  satisfies the energy inequality  $\frac{1}{2} \|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u\|_{L^2}^2 \leq \frac{1}{2} \|u_0\|_{L^2}^2$ ; such solutions exist globally [9]. A *strong solution* is a solution in  $C([0, T); H^s)$ ,  $s > 5/2$ ; these exist locally in time, are smooth for  $t > 0$  while they persist, and can be continued as long as  $\|u(t)\|_{H^s}$  stays bounded (Kato’s theory; see [1]). We will use the commutator (tame) estimate of Kato–Ponce [7]: for  $s > 5/2$ , along strong solutions,

$$\frac{d}{dt} \|u(t)\|_{H^s}^2 \leq C_s \|\nabla u(t)\|_{L^\infty} \|u(t)\|_{H^s}^2, \quad (5)$$

the dissipative term having a favorable sign. Finally we recall Foias–Temam Gevrey regularity [6]: if  $u$  is a strong solution on  $(0, T)$ , then for each compact  $[t_0, t_1] \subset (0, T)$  there are  $\tau > 0$ ,  $C > 0$  with

$$\|\Delta_j u(t)\|_{L^2} \leq C e^{-\tau 2^j}, \quad t \in [t_0, t_1], \quad j \geq 0. \quad (6)$$

## 3 Transfer operators and the spectral determinant

**Definition 3.1** (Shell spaces and transfer operators). *Let  $V_j \subset L_\sigma^2(\mathbf{T}^3)$  be the (finite-dimensional) space of mean-free, divergence-free fields with Fourier support in  $\text{supp } \varphi(2^{-j} \cdot)$ ; thus*

$$d_j := \dim V_j \leq C 2^{3j}. \quad (7)$$

*For a solution  $u$  define  $A_j(t): V_j \rightarrow V_j$  by*

$$A_j(t)v = \Delta_j \mathbf{P}[(\tilde{\Delta}_j u(t) \cdot \nabla)v].$$

*$A_j(t)$  encodes the scale-local part of the nonlinear energy transfer through the  $j$ -th shell. (Non-local interactions are deliberately not encoded; see Remark 3.6.)*

Because  $d_j < \infty$ , the determinant below is the classical finite-dimensional determinant, and no regularization is required; working on the torus is what makes this possible.

**Definition 3.2** (Spectral determinant and completed function). *For  $s \in \mathbf{C}$  and  $t \geq 0$  set, whenever the product converges,*

$$D_u(s, t) = \prod_{j \geq 1} \det(I - A_j(t) \lambda_j^{-s}), \quad \Lambda_u(s, t) = \Gamma_\infty(s) D_u(s, t)^{-1}, \quad \Gamma_\infty(s) = \pi^{-s/2} \Gamma(s/2).$$

*Remark 3.3* (Two normalization conventions). (a) *The product starts at  $j = 1$ .* Since  $\lambda_0 = 2^0 = 1$ , the shell-0 factor  $\det(I - A_0(t) \lambda_0^{-s}) = \det(I - A_0(t))$  is constant in  $s$ : it carries no analytic information, and it vanishes precisely when  $1 \in \text{spec } A_0(t)$ , which can occur at isolated times and would then force  $D_u(\cdot, t) \equiv 0$ , leaving  $\Lambda_u(\cdot, t)$  undefined. Omitting the  $j = 0$  factor loses nothing and removes this spurious degeneracy. Consistently, the pole-location formula (9) below divides by  $j \log 2$  and is stated for  $j \geq 1$  only.

(b) *The archimedean factor is conventional.* We take  $\Gamma_\infty(s) = \pi^{-s/2} \Gamma(s/2)$  for definiteness, in imitation of the Riemann  $\xi$ -function. Unlike the archimedean local factor in Tate's theory [13], which is *computed* from a local zeta integral,  $\Gamma_\infty$  here is not derived from any structure of the dynamics. Remark 6.2 makes precise which of our conclusions depend on this choice and in what way; the answer, in brief, is that the nonexistence of a functional equation does not (Proposition 6.1(ii)).

**Lemma 3.4** (Operator bounds). *For every  $t$  at which  $u(t) \in L^2$ ,*

$$\|A_j(t)\|_{V_j \rightarrow V_j} \leq C 2^j \|\tilde{\Delta}_j u(t)\|_{L^\infty} \leq C \min\left(2^{5j/2} \|u(t)\|_{L^2}, \|u(t)\|_{\dot{B}_{\infty, \infty}^1}\right). \quad (8)$$

*Proof.* For  $v \in V_j$ , using that  $\mathbf{P}$  and  $\Delta_j$  are  $L^2$ -orthogonal projections and (3),

$$\|A_j v\|_{L^2} \leq \|(\tilde{\Delta}_j u \cdot \nabla) v\|_{L^2} \leq \|\tilde{\Delta}_j u\|_{L^\infty} \|\nabla v\|_{L^2} \leq C 2^j \|\tilde{\Delta}_j u\|_{L^\infty} \|v\|_{L^2}.$$

The first alternative in (8) follows from  $\|\tilde{\Delta}_j u\|_{L^\infty} \leq C 2^{3j/2} \|u\|_{L^2}$  (Bernstein), the second from  $\|\tilde{\Delta}_j u\|_{L^\infty} \leq C 2^{-j} \|u\|_{\dot{B}_{\infty, \infty}^1}$  by definition of the Besov norm.  $\square$

**Proposition 3.5** (Unconditional half-plane of holomorphy). *Let  $u$  be a Leray–Hopf solution and  $t \geq 0$ . Then the product defining  $D_u(s, t)$  converges absolutely and locally uniformly on  $\{\text{Re } s > 11/2\}$ , where  $D_u(\cdot, t)$  is holomorphic; hence  $\Lambda_u(\cdot, t)$  is meromorphic there. Moreover every zero of  $D_u(\cdot, t)$  (pole of  $\Lambda_u$ ) in that region — and, wherever  $D_u$  extends, in general — lies in*

$$\left\{ s : \text{Re } s \leq \frac{5}{2} + \frac{\log(C \|u(t)\|_{L^2})}{j \log 2} \text{ for some } j \geq 1 \right\},$$

*so the pole abscissae accumulate at most at  $5/2$ .*

*Proof.* Write  $B_j = A_j(t) \lambda_j^{-s}$  and  $\sigma = \text{Re } s$ . By Lemma 3.4,  $\|B_j\| \leq C 2^{j(5/2 - \sigma)} \|u(t)\|_{L^2}$ , which is  $\leq \frac{1}{2}$  for  $j \geq j_0(\sigma)$  locally uniformly in  $s$ . For a  $d$ -dimensional operator  $B$  with  $\|B\| \leq \frac{1}{2}$ , let  $\beta_1, \dots, \beta_d$  be its eigenvalues (with multiplicity); each satisfies  $|\beta_i| \leq \rho(B) \leq \|B\| \leq \frac{1}{2}$ , and since  $|\log(1 - z)| \leq 2|z|$  for  $|z| \leq \frac{1}{2}$  (principal branch),

$$\left| \sum_{i=1}^d \log(1 - \beta_i) \right| \leq \sum_{i=1}^d |\log(1 - \beta_i)| \leq 2 \sum_i |\beta_i| \leq 2d \|B\|,$$

and  $\sum_i \log(1 - \beta_i)$  is a logarithm of  $\det(I - B) = \prod_i (1 - \beta_i)$ . Hence by (7)

$$\sum_{j \geq j_0} \left| \sum_i \log(1 - \beta_i(B_j)) \right| \leq C \|u(t)\|_{L^2} \sum_{j \geq j_0} 2^{3j} 2^{j(5/2 - \sigma)} < \infty \quad \text{iff} \quad \sigma > \frac{11}{2},$$

locally uniformly; absolute and locally uniform convergence of the product and holomorphy follow. For the pole location:  $\det(I - A_j \lambda_j^{-s}) = 0$  iff  $\lambda_j^s$  is a nonzero eigenvalue  $\alpha$  of  $A_j(t)$ , i.e. (using  $j \geq 1$ , so  $\log \lambda_j = j \log 2 > 0$ )

$$s = \frac{\log |\alpha|}{j \log 2} + i \frac{\arg \alpha + 2\pi m}{j \log 2}, \quad m \in \mathbf{Z}, j \geq 1, \quad (9)$$

and  $|\alpha| \leq \|A_j(t)\| \leq C 2^{5j/2} \|u(t)\|_{L^2}$  gives  $\operatorname{Re} s \leq \frac{5}{2} + \frac{\log(C\|u\|_{L^2})}{j \log 2}$ .  $\square$

*Remark 3.6.*  $A_j$  retains only quasi-local (in scale) interactions, as in shell models [3]. This is a modeling choice: the full nonlinearity also transfers energy between distant shells. Nothing below claims that  $\Lambda_u$  determines the solution; it is a diagnostic attached to it. We have likewise made no attempt to optimize the abscissa 11/2 of Proposition 3.5; trace-norm estimates on  $A_j$  would improve it, but nothing in the sequel depends on its value.

## 4 Meromorphic continuation from smoothing, not symmetry

**Theorem 4.1** (Entire determinant along strong solutions). *Let  $u$  be a strong solution of (1) on  $(0, T)$ . Then for every  $t \in (0, T)$  the product defining  $D_u(s, t)$  converges locally uniformly on all of  $\mathbf{C}$ ;  $D_u(\cdot, t)$  is entire, and  $\Lambda_u(\cdot, t)$  is meromorphic on  $\mathbf{C}$ , with pole set*

$$\mathcal{P}(t) = \{-2m : m \geq 0\} \cup \{s \text{ as in (9)} : \alpha \in \operatorname{spec} A_j(t) \setminus \{0\}, j \geq 1\},$$

*the first set coming from  $\Gamma_\infty$ . (No cancellation can occur in this union:  $\Gamma_\infty$  is zero-free, and  $1/D_u$  has no zeros because  $D_u$  is entire; where the two sets meet, orders add.) Moreover, with  $\tau = \tau(t) > 0$  the Gevrey radius in (6), all but finitely many poles satisfy*

$$\operatorname{Re} s \leq \frac{5}{2} - \frac{\tau 2^j}{j \log 2} \xrightarrow{j \rightarrow \infty} -\infty. \quad (10)$$

*Proof.* By (6) and Bernstein,  $\|\tilde{\Delta}_j u(t)\|_{L^\infty} \leq C 2^{3j/2} e^{-\tau 2^j}$ , hence by Lemma 3.4  $\|A_j(t)\| \leq C 2^{5j/2} e^{-\tau 2^j}$ . For any compact  $K \subset \mathbf{C}$  and  $s \in K$ ,  $\|A_j(t) \lambda_j^{-s}\| \leq C 2^{j(5/2 + \sigma_K)} e^{-\tau 2^j} \rightarrow 0$  superexponentially, where  $\sigma_K = \max_K |\operatorname{Re} s|$ ; the estimate in the proof of Proposition 3.5 then gives absolute, locally uniform convergence of  $\sum_{j \geq 1} \log \det(I - A_j \lambda_j^{-s})$  on  $K$ . An absolutely and locally uniformly convergent product of entire functions (each  $\det(I - A_j \lambda_j^{-s})$  is a polynomial in  $\lambda_j^{-s} = e^{-sj \log 2}$ , hence entire) is entire. Zeros of the product are exactly the union of zeros of the factors, giving  $\mathcal{P}(t)$  via (9); the bound (10) follows from  $|\alpha| \leq C 2^{5j/2} e^{-\tau 2^j}$ .  $\square$

*Remark 4.2* (What the continuation does and does not encode). By analogy with Hecke theory one might hope to obtain the continuation of  $\Lambda_u$  from a functional equation and then extract regularity information from the location of its poles. Theorem 4.1 shows the continuation is real but has an entirely different, and unconditional, source: *parabolic smoothing*. Spatial analyticity of the flow (Foias–Temam) forces the transfer amplitudes to decay so fast that the Euler-type product converges everywhere. Consequently, continuation of  $\Lambda_u$  carries no information about global regularity beyond what analyticity already encodes: if the strong solution has maximal existence time  $T^* < \infty$ , then as  $t \uparrow T^*$  the Gevrey radius  $\tau(t)$  may degenerate to 0 and the right-most pole front in (10) may escape to  $+\infty$ . We emphasize that Lemma 3.4 provides only *upper* bounds on  $\|A_j\|$ ; the converse implication (pole confinement  $\Rightarrow$  regularity) would require lower bounds relating  $\|\Delta_j u\|_{L^\infty}$  to  $\operatorname{spec} A_j$ , which are not available in general because the  $A_j$  are far from normal. This gap is isolated as Question 7.7 and probed numerically in §8.

## 5 Temperedness: the exact strength of pole confinement

We now identify the quantitative content of the Ramanujan-type (“tempered”) normalization in which the poles of  $\Lambda_u$  are confined near  $\operatorname{Re} s = 0$ , the analogue of local eigenvalues on the unit circle. By Lemma 3.4, such confinement — indeed the stronger statement  $\sup_j \|A_j(t)\| < \infty$  — is implied by a uniform bound on  $\|u(t)\|_{\dot{B}_{\infty,\infty}^1} = \sup_j 2^j \|\Delta_j u(t)\|_{L^\infty}$ . We adopt the latter as the hypothesis; it is the statement about the flow that actually does the work, and §6 shows nothing weaker phrased through  $\operatorname{spec} A_j$  alone could suffice, for lack of lower bounds.

**Assumption 5.1** (Tempered spectrum, quantitative form).  *$u$  is a strong solution on  $(0, T)$  with*

$$M_T := \int_0^T \|u(t)\|_{\dot{B}_{\infty,\infty}^1} dt < \infty.$$

By (4) this is equivalent to  $\int_0^T \|\omega(t)\|_{\dot{B}_{\infty,\infty}^0} dt < \infty$ , a vorticity condition one full derivative weaker than  $\int_0^T \|\omega\|_{L^\infty} dt < \infty$  would suggest at first sight, the inclusion  $L^\infty \subset \dot{B}_{\infty,\infty}^0$  being strict; the theorem below is, accordingly, a known type of result [8]. We include a complete proof to keep the paper self-contained and to make plain that nothing beyond classical harmonic analysis enters.

**Lemma 5.2** (Logarithmic interpolation). *Let  $s > 5/2$ . There is  $C = C(s)$  such that for all mean-free  $u \in H^s$  with  $\nabla \cdot u = 0$  and  $\omega = \nabla \times u$ ,*

$$\|\nabla u\|_{L^\infty} \leq C \left( 1 + \|\omega\|_{\dot{B}_{\infty,\infty}^0} \log(e + \|u\|_{H^s}) \right).$$

*Proof.* Fix  $J \geq 1$  to be chosen. Since on each shell  $\nabla u = T_j \omega$  with  $T_j$  a multiplier of degree 0 (uniformly bounded on  $L^\infty$  over  $j$ ),

$$\|\nabla u\|_{L^\infty} \leq \sum_{j \leq J} \|\Delta_j \nabla u\|_{L^\infty} + \sum_{j > J} \|\Delta_j \nabla u\|_{L^\infty} \leq C(J+1) \|\omega\|_{\dot{B}_{\infty,\infty}^0} + C \sum_{j > J} 2^{j(5/2-s)} \|u\|_{H^s},$$

using  $\|\Delta_j \nabla u\|_{L^\infty} \leq C 2^{3j/2} \|\Delta_j \nabla u\|_{L^2} \leq C 2^{5j/2} \|\Delta_j u\|_{L^2} \leq C 2^{j(5/2-s)} \|u\|_{H^s}$  in the tail. The tail sums to  $C 2^{J(5/2-s)} \|u\|_{H^s}$ ; choosing  $J = \lceil \frac{\log(e + \|u\|_{H^s})}{(s-5/2) \log 2} \rceil$  makes it  $\leq C$  and gives the claim.  $\square$

**Theorem 5.3** (Temperedness implies regularity). *Under Assumption 5.1,  $\sup_{0 < t < T} \|u(t)\|_{H^s} < \infty$  for  $s > 5/2$ ; consequently the solution extends smoothly past  $T$ . In particular, if Assumption 5.1 holds for every  $T < \infty$ , the solution is globally smooth.*

*Proof.* Let  $y(t) = \log(e + \|u(t)\|_{H^s}^2)$ . By (5) and Lemma 5.2, using (4) in the form  $\|\omega\|_{\dot{B}_{\infty,\infty}^0} \leq C \|u\|_{\dot{B}_{\infty,\infty}^1}$ , and noting  $\log(e + x) \leq 2 \log(e + x^2)$  for all  $x \geq 0$  (so that  $\log(e + \|u\|_{H^s}) \leq 2y$ ),

$$y'(t) \leq C \|\nabla u\|_{L^\infty} \leq C \left( 1 + 2 \|u(t)\|_{\dot{B}_{\infty,\infty}^1} y(t) \right) \leq C (1 + \|u(t)\|_{\dot{B}_{\infty,\infty}^1}) y(t),$$

since  $y \geq 1$ . Grönwall gives  $y(t) \leq y(0^+) \exp(C(t + M_T))$  for  $t < T$ , i.e. a (double-exponential) a priori bound on  $\|u(t)\|_{H^s}$ , uniform on  $(0, T)$ . By the continuation criterion for strong solutions,  $u$  extends past  $T$ ; smoothness for positive times is standard parabolic bootstrap.  $\square$

**Proposition 5.4** (Regularity implies temperedness). *Conversely, if  $u$  is a strong solution on  $[0, T]$  (in particular  $\sup_{[0, T]} \|u\|_{H^s} < \infty$  for some  $s > 5/2$ ), then Assumption 5.1 holds on  $(0, T)$ :*

$$\int_0^T \|u\|_{\dot{B}_{\infty, \infty}^1} dt \leq T \sup_{[0, T]} \|u\|_{\dot{B}_{\infty, \infty}^1} \leq CT \sup_{[0, T]} \|\nabla u\|_{L^\infty} \leq C_s T \sup_{[0, T]} \|u\|_{H^s} < \infty.$$

*Proof.*  $2^j \|\Delta_j u\|_{L^\infty} \leq C \|\Delta_j \nabla u\|_{L^\infty} \leq C \|\nabla u\|_{L^\infty}$  by Bernstein on shells, and  $H^s \hookrightarrow W^{1, \infty}$  for  $s > 5/2$ .  $\square$

**Corollary 5.5** (The dictionary is exact — and therefore not a reduction). *For strong solutions, Assumption 5.1 on  $(0, T)$  and smooth extendability past  $T$  are equivalent. The spectral-determinant framework therefore converts the regularity problem into an exactly equally hard statement about shell amplitudes; it does not weaken it. Any approach that postulates a tempered normalization of the transfer spectra is postulating regularity itself.*

## 6 Nonexistence of a functional equation

What survives of the scale duality  $\lambda \mapsto 1/\lambda$  is only the elementary Mellin identity  $\int_0^\infty f(\lambda) \lambda^s \frac{d\lambda}{\lambda} = \int_0^\infty f(1/\lambda) \lambda^{-s} \frac{d\lambda}{\lambda}$ , which constrains nothing unless the data are themselves symmetric. For Navier–Stokes they are not: dissipation  $\nu \lambda^2$  maps to  $\nu \lambda^{-2}$  under  $\lambda \mapsto 1/\lambda$ , and the energy cascade is directional. The following makes the failure precise: in exactly the tempered regime that the analogy with  $L$ -functions would posit, the functional equation is impossible — both for the completed function and, whenever the dynamical pole set is nonempty, for the bare determinant itself.

**Proposition 6.1** (No functional equation in the tempered regime). *Let  $t$  be such that  $D_u(\cdot, t)$  is entire (e.g. any  $t$  in the lifespan of a strong solution, Theorem 4.1), and suppose the flow is tempered at time  $t$  in the spectral sense that*

$$\sigma_M(t) := \sup_{j \geq 1} \frac{\log^+ \rho(A_j(t))}{j \log 2} < \frac{1}{2},$$

where  $\rho$  denotes spectral radius (the unit-circle normalization corresponds to  $\sigma_M = 0$ ). Then:

(i) *there is no function  $\varepsilon(t)$  with  $|\varepsilon(t)| = 1$  such that*

$$\Lambda_u(s, t) = \varepsilon(t) \Lambda_u(1 - s, t) \quad \text{on } \mathbf{C};$$

(ii) *if moreover some  $A_j(t)$ ,  $j \geq 1$ , has a nonzero eigenvalue — equivalently, if the dynamical pole set  $\mathcal{P}_{\text{dyn}}(t)$  of Theorem 4.1 is nonempty — then likewise there is no  $\varepsilon(t)$  with  $|\varepsilon(t)| = 1$  such that*

$$D_u(s, t) = \varepsilon(t) D_u(1 - s, t) \quad \text{on } \mathbf{C}.$$

*Proof.* (i) Suppose the identity holds. Poles are then symmetric: writing  $\mathcal{P}$  for the pole set of  $\Lambda_u(\cdot, t)$ , we get  $\mathcal{P} = 1 - \mathcal{P}$ . By Theorem 4.1,  $\mathcal{P} = \{-2m : m \geq 0\} \cup \mathcal{P}_{\text{dyn}}$  where, by (9) and the definition of  $\sigma_M$ , every  $s \in \mathcal{P}_{\text{dyn}}$  has  $\text{Re } s \leq \sigma_M < \frac{1}{2}$ ; also every pole of  $\Gamma_\infty$  has  $\text{Re } s \leq 0 < \frac{1}{2}$ . Hence  $\mathcal{P} \subset \{\text{Re } s < \frac{1}{2}\}$ , while  $1 - \mathcal{P} \subset \{\text{Re } s > \frac{1}{2}\}$ . Since  $s = 0 \in \mathcal{P}$  (a pole of  $\Gamma_\infty$ ; it is not cancelled, because  $1/D_u$  has poles, never zeros,  $D_u$  being entire),  $\mathcal{P} \neq \emptyset$ , so  $\mathcal{P} = 1 - \mathcal{P}$  is impossible.



(ii) Suppose the identity holds and let  $\mathcal{Z}$  denote the zero set of  $D_u(\cdot, t)$ ; the identity forces  $\mathcal{Z} = 1 - \mathcal{Z}$ . By Theorem 4.1,  $\mathcal{Z} = \mathcal{P}_{\text{dyn}}(t)$ , which by hypothesis is nonempty and, by (9) and the definition of  $\sigma_M$ , is contained in  $\{\text{Re } s \leq \sigma_M < \frac{1}{2}\}$ . Then  $\mathcal{Z} \subset \{\text{Re } s < \frac{1}{2}\}$  and  $1 - \mathcal{Z} \subset \{\text{Re } s > \frac{1}{2}\}$  with  $\mathcal{Z} \neq \emptyset$ : contradiction.  $\square$

*Remark 6.2* (On the role of  $\Gamma_\infty$ ; nilpotency). Two comments on Proposition 6.1. First, in part (i) the nonemptiness of the pole set — the step that makes the symmetry argument bite — is supplied entirely by the archimedean factor, which (Remark 3.3(b)) is a convention rather than an output of the dynamics; part (i) should therefore be read as the statement that no unimodular constant makes *this particular completion* self-dual. Part (ii) removes the convention: whenever the dynamics produces any nonzero transfer eigenvalue, the bare determinant itself admits no functional equation — and then no completion  $G/D_u$  by an archimedean factor with its own reflection symmetry  $G(s) = \varepsilon_G G(1-s)$ ,  $|\varepsilon_G| = 1$ , can admit one either, since  $G\Lambda$ -type identities for  $G/D_u$  and for  $G$  together force one for  $D_u$ . Second, the hypothesis of part (ii) is a genuine caveat, not a formality: if every  $A_j(t)$  were nilpotent, then  $\det(I - A_j \lambda_j^{-s}) \equiv 1$  identically in  $s$ , so  $D_u(\cdot, t) \equiv 1$  and the functional equation would hold trivially and vacuously. Nilpotency of transfer operators is not a fanciful possibility — the per-shell Jacobian blocks of the Sabra model vanish *identically* (Lemma 7.1) — which is precisely why the empirical lower bounds of §8, showing the windowed Sabra spectra are far from nilpotent, carry content.

*Remark 6.3.* One can of course *define* a symmetrized object  $\tilde{\Lambda}(s) = \Lambda_u(s)\Lambda_u(1-s)$  that satisfies a functional equation by fiat; it then carries no dynamical information. A functional equation is meaningful only as a theorem about the dynamics — in Tate’s setting [13] it is Poisson summation on the adèles, a genuine self-duality of the underlying arithmetic. The Navier–Stokes cascade possesses no analogous self-duality: there is no multiplicative structure making the local factors interact (no Hecke relations, no Euler product in the arithmetic sense), and no analogue of the product formula — energy is *additive* across shells ( $\sum_j \|\Delta_j u\|_{L^2}^2 \approx \|u\|_{L^2}^2$ ), whereas the adelic norm relation  $\prod_v |x|_v = 1$  is multiplicative. This is where the analogy with  $L$ -functions ends.

## 7 The Sabra shell model: explicit transfer matrices

### 7.1 The model and its realification

The viscous Sabra model [10, 3, 4] evolves  $u_n(t) \in \mathbf{C}$ ,  $0 \leq n \leq N-1$ ,  $N \leq \infty$  (with  $u_m \equiv 0$  for  $m < 0$  or  $m \geq N$ ), with wavenumbers  $k_n = k_0 \mu^n$ ,  $\mu > 1$ ; throughout we assume  $k_0 \geq 1$ , so that  $k_n > 1$  for all  $n \geq 1$  (as in the numerics of §8, where  $k_0 = 1$ ,  $\mu = 2$ ):

$$\dot{u}_n = B_n(u) - \nu k_n^2 u_n + f_n, \quad B_n(u) = i(ak_{n+1}u_{n+2}\bar{u}_{n+1} + bk_n u_{n+1}\bar{u}_{n-1} - ck_{n-1}u_{n-1}u_{n-2}), \quad (11)$$

with  $a + b + c = 0$ , which is equivalent to conservation of the energy  $E = \frac{1}{2} \sum_n |u_n|^2$  by the inviscid, unforced flow.

Because  $B_n$  involves complex conjugates, its derivative is  $\mathbf{R}$ -linear but not  $\mathbf{C}$ -linear, and the natural transfer matrices live on the realification  $\mathbf{C} \cong \mathbf{R}^2$ ,  $u = x + iy$ . For  $w \in \mathbf{C}$  let

$$\mathbf{R}(w) = \begin{pmatrix} \text{Re } w & -\text{Im } w \\ \text{Im } w & \text{Re } w \end{pmatrix}, \quad \mathbf{K}(w) = \begin{pmatrix} \text{Re } w & \text{Im } w \\ \text{Im } w & -\text{Re } w \end{pmatrix}$$

be the real  $2 \times 2$  matrices of the maps  $v \mapsto wv$  ( $\mathbf{C}$ -linear) and  $v \mapsto w\bar{v}$  (antilinear). Both satisfy  $\|\mathbf{R}(w)\| = \|\mathbf{K}(w)\| = |w|$ , being  $|w|$  times a rotation, respectively a reflection.

**Lemma 7.1** (Explicit Jacobian). *Let  $J(t)$  be the realified Jacobian of  $u \mapsto B(u)$  at  $u(t)$ , with  $2 \times 2$  blocks  $J_{n,m} = \partial B_n / \partial u_m$ . Then  $J_{n,m} = 0$  for  $|n - m| \geq 3$  and for  $m = n$ , and*

$$\begin{aligned} J_{n,n+2} &= R(i a k_{n+1} \bar{u}_{n+1}), & J_{n,n+1} &= K(i a k_{n+1} u_{n+2}) + R(i b k_n \bar{u}_{n-1}), \\ J_{n,n-2} &= R(-i c k_{n-1} u_{n-1}), & J_{n,n-1} &= K(i b k_n u_{n+1}) + R(-i c k_{n-1} u_{n-2}). \end{aligned}$$

*Proof.* Differentiate the three monomials in (11) term by term: a monomial  $w u_m$  contributes  $R(w)$  to  $J_{n,m}$  and a monomial  $w \bar{u}_m$  contributes  $K(w)$ , with  $w$  the (frozen) coefficient. The first term  $i a k_{n+1} u_{n+2} \bar{u}_{n+1}$  contributes  $R(i a k_{n+1} \bar{u}_{n+1})$  at  $m = n+2$  and  $K(i a k_{n+1} u_{n+2})$  at  $m = n+1$ ; the second,  $i b k_n u_{n+1} \bar{u}_{n-1}$ , contributes  $R(i b k_n \bar{u}_{n-1})$  at  $m = n+1$  and  $K(i b k_n u_{n+1})$  at  $m = n-1$ ; the third,  $-i c k_{n-1} u_{n-1} u_{n-2}$ , contributes  $R(-i c k_{n-1} u_{n-2})$  at  $m = n-1$  and  $R(-i c k_{n-1} u_{n-1})$  at  $m = n-2$ . No monomial contains  $u_n$  or  $\bar{u}_n$ , so  $J_{n,n} = 0$ . (In the companion code the formulas are additionally verified against a central finite-difference Jacobian, with relative error  $6 \times 10^{-13}$ ; §8.)  $\square$

**Remark 7.2** (The diagonal vanishes: windows are forced).  $J_{n,n} = 0$  says that the within-shell transfer operator — the direct analogue of Definition 3.1, which is nontrivial on  $\mathbf{T}^3$  because a Littlewood–Paley shell contains many mutually advecting modes — vanishes identically in any shell model whose triads couple only distinct shells. The minimal meaningful local factors in a shell model are therefore compressions of  $J$  to *windows* of shells, as follows. This is a structural asymmetry between the shell-model and Navier–Stokes settings worth keeping in view when transferring intuition between them.

**Definition 7.3** (Windowed transfer operators). *For a window radius  $r \geq 1$  and  $r \leq n \leq N - 1 - r$ , let  $W_n^{(r)} = \bigoplus_{|m-n| \leq r} \mathbf{R}_m^2$  (dimension  $2(2r+1)$ ) and let*

$$A_n^{(r)}(t) := \Pi_n^{(r)} J(t) \Pi_n^{(r)} \Big|_{W_n^{(r)}},$$

*the compression of the Jacobian of Lemma 7.1 to the window. The spectral determinant is  $D_u(s, t) = \prod_{r \leq n \leq N-1-r} \det(I - A_n^{(r)}(t) k_n^{-s})$  and  $\Lambda_u = \Gamma_\infty / D_u$  as before. Since  $n \geq r \geq 1$  and  $k_0 \geq 1$ , every factor has  $k_n > 1$ , so  $\log k_n > 0$  and the analogue of the pole-location formula (9) is well defined; the degeneracy of Remark 3.3(a) does not arise. (Windows overlap;  $D_u$  is a diagnostic attached to the solution, not a factorization of  $\det(I - Jk^{-s})$ .)*

**Lemma 7.4** (Norm bound). *With  $\kappa = \max(|a|\mu, |b|, |c|/\mu)$  and  $U_n(t) = \max_{|m-n| \leq r+2} |u_m(t)|$ ,*

$$\|A_n^{(r)}(t)\| \leq 8 \kappa \mu^r k_n U_n(t).$$

*Proof.* Each nonzero block of  $A_n^{(r)}$  is, by Lemma 7.1, a sum of at most two matrices of the form  $R(w)$  or  $K(w)$  with  $|w| \leq \kappa k_p U_n \leq \kappa \mu^r k_n U_n$  for the row index  $p$  in the window; hence each block has norm  $\leq 2\kappa \mu^r k_n U_n$ , and each block-row and block-column contains at most 4 nonzero blocks. For a block matrix,  $\|A\| \leq (\max_p \sum_q \|A_{pq}\|)^{1/2} (\max_q \sum_p \|A_{pq}\|)^{1/2} \leq 4 \cdot 2\kappa \mu^r k_n U_n$ .  $\square$

**Theorem 7.5** (Unconditional meromorphic continuation for Sabra). *Let  $\nu > 0$ ,  $u(0) \in \ell^2 =: H$ , and let  $f$  be supported on finitely many shells (or Gevrey-decaying);  $N \leq \infty$ . Write  $V := \{v : \|v\|_V^2 = \sum_n k_n^2 |v_n|^2 < \infty\}$ . Then:*

- (i) *the solution is unique and global, and for every  $t_0 > 0$  there are  $C, \tau > 0$  with  $|u_n(t)| \leq C e^{-\tau k_n}$  for all  $t \geq t_0$ ,  $n \geq 0$  (Constantin–Levant–Titi [4]: global well-posedness and Gevrey regularity for the viscous model);*

(ii) for every  $0 < t_0 \leq T < \infty$ , the shell-model form of Assumption 5.1 holds on  $[t_0, T]$ :  $\int_{t_0}^T \sup_n k_n |u_n| dt < \infty$ . If moreover  $u(0) \in V$ , the bound holds with  $t_0 = 0$ , i.e. on every  $[0, T]$ , unconditionally. (For data merely in  $H$  the integrand need not be finite at  $t = 0$ ; nothing in (iii) requires it.)

(iii) for every  $t > 0$  and every  $r \geq 1$ ,  $D_u(\cdot, t)$  is entire and  $\Lambda_u(\cdot, t)$  is meromorphic on  $\mathbf{C}$ , with dynamical poles given by (9) (with  $\lambda_j$  replaced by  $k_n$ ,  $j \log 2$  by  $\log k_n$ , and  $A_j$  by  $A_n^{(r)}$ ), all but finitely many of which satisfy

$$\operatorname{Re} s \leq 1 + \frac{\log C'}{\log k_n} - \frac{\tau'' k_n}{\log k_n} \xrightarrow{n \rightarrow \infty} -\infty, \quad \tau'' := \tau \mu^{-(r+2)}.$$

*Proof.* (i) is [4]. (ii): for  $t \geq t_0$ , by (i),  $k_n |u_n(t)| \leq C k_n e^{-\tau k_n}$  is bounded uniformly in  $n$  and  $t$ , so the integral over  $[t_0, T]$  is finite. If  $u(0) \in V$ , then  $u \in C([0, T]; V)$  by the global well-posedness of strong solutions with  $V$  data [4], and  $\sup_n k_n |u_n(t)| \leq \|u(t)\|_V$  is bounded on  $[0, T]$ , so the integral is finite down to  $t_0 = 0$ . (iii): by Lemma 7.4 and (i), for  $t \geq t_0 > 0$ ,

$$\|A_n^{(r)}(t)\| \leq 8\kappa \mu^r k_n C e^{-\tau k_n - r} = C' k_n e^{-\tau'' k_n}, \quad \tau'' = \tau \mu^{-(r+2)},$$

which decays superexponentially in  $n$ ; each factor  $\det(I - A_n^{(r)} k_n^{-s})$  is a polynomial in  $k_n^{-s}$ , hence entire, and the argument of Theorem 4.1 applies verbatim (the pole abscissa bound follows from  $|\alpha| \leq C' k_n e^{-\tau'' k_n}$  and  $\log k_n > 0$ ).  $\square$

The Sabra model thus plays the role of a solvable sandbox: every analytic feature of the determinant is a theorem there, *because* global regularity was proved first, by energy methods [4]. The formalism inherits regularity; it does not produce it — the same ordering that Corollary 5.5 establishes for the true equations. The following triviality is the dictionary used in §8.

**Lemma 7.6** (Pole abscissa reads off scaling). *Fix  $t$  and suppose that for  $n_1 \leq n \leq n_2$  (with  $n_1 \geq r \geq 1$ , so  $k_n > 1$ ) there are  $0 < c \leq C$  and  $\beta \in \mathbf{R}$  with  $c k_n^\beta \leq \rho(A_n^{(r)}(t)) \leq C k_n^\beta$ . Then for each such  $n$  the outermost dynamical pole contributed by shell  $n$  has real part  $\sigma_n$  with*

$$|\sigma_n - \beta| \leq \frac{\max(\log C, \log(1/c))}{\log k_n}.$$

*Proof.* Immediate from (9): the outermost pole from shell  $n$  has  $\sigma_n = \log \rho(A_n^{(r)}) / \log k_n$ .  $\square$

If the cascade obeys  $|u_n| \sim k_n^{-h}$  in a typical sense, then Lemma 7.4 gives  $\rho(A_n) \leq C k_n^{1-h}$ ; whether  $\rho$  also obeys the matching *lower* bound is exactly the question of whether the spectrum of the (non-normal) transfer matrices sees the amplitudes. We record both forms:

**Question 7.7** (Spectral lower bound). *Do there exist  $c, \theta > 0$  such that, along solutions,  $\rho(A_n^{(r)}(t)) \geq c (k_n |u_n(t)|)^\theta$  (Sabra), respectively  $\sup_{|i-j| \leq N_0} \rho(A_i(t)) \geq c (2^j \|\Delta_j u(t)\|_{L^\infty})^\theta$  (Navier–Stokes)?*

For Kolmogorov scaling  $h = 1/3$  the predicted pole front is  $\beta = 1 - h = 2/3$ ; intermittency shifts the *typical* (mean-logarithm) amplitude exponent, and it is against that exponent that the front should be tested. We now do so.

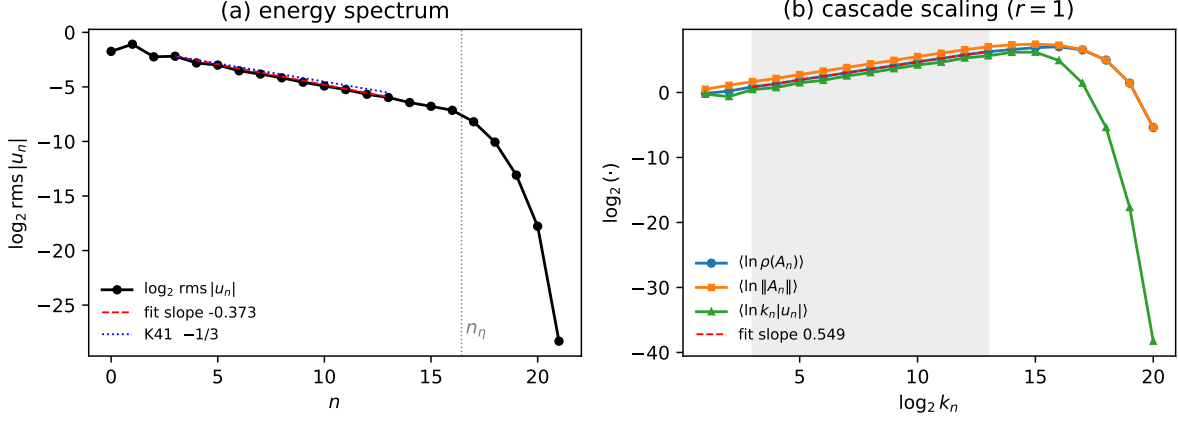


Figure 1: (a) Time-averaged spectrum, fitted slope  $-0.373$  (Kolmogorov:  $-1/3$ ), with the dissipative cutoff at  $n_\eta \approx 16.4$ . (b) Cascade scaling of the mean-log spectral radius  $\langle \ln \rho(A_n^{(1)}) \rangle$ , operator norm  $\langle \ln \|A_n^{(1)}\| \rangle$ , and local amplitude  $\langle \ln k_n |u_n| \rangle$ ; the three run parallel through the inertial range (shaded), with fitted pole-front slope  $0.549$ , and all plunge in the dissipation range — the Gevrey mechanism of Theorem 7.5(iii) made visible.

## 8 Numerical pole structure of $\Lambda_u$ in Sabra turbulence

### 8.1 Setup and verification

We integrate (11) with  $N = 22$  shells,  $\mu = 2$ ,  $k_0 = 1$ , the standard energy-conserving coefficients  $a = 1$ ,  $b = c = -\frac{1}{2}$ , viscosity  $\nu = 10^{-7}$ , and constant forcing  $f_1 = 0.1(1 + i)$  on shell  $n = 1$ , using an integrating-factor RK4 scheme (viscous term integrated exactly) with  $\Delta t = 10^{-4}$ , a discarded transient of 60 time units from a Kolmogorov-shaped random-phase initial condition, and a sampling window of 200 time units ( $\approx 10^2$  large-eddy turnovers) from which 1000 snapshots are retained. Four independent correctness checks are enforced programmatically:

- C1. *Energy conservation.* For the inviscid, unforced model (integrated with plain RK4 over a window short enough to precede the inviscid blow-up), the relative energy drift is  $1.1 \times 10^{-14}$ .
- C2. *Jacobian.* The closed-form Jacobian of Lemma 7.1 agrees with a central finite-difference Jacobian of the realified nonlinearity to relative error  $6.2 \times 10^{-13}$ .
- C3. *Stationarity.* In the sampling window, mean energy input  $\langle \text{Re} \sum f_n \bar{u}_n \rangle = 0.0606$  balances mean dissipation  $\langle \nu \sum k_n^2 |u_n|^2 \rangle = 0.0609$  to 0.6%; thus  $\varepsilon \approx 0.061$  and the Kolmogorov shell is  $n_\eta = \log_2(\varepsilon/\nu^3)^{1/4} \approx 16.4$ .
- C4. *Spectrum.* The rms amplitude scales as  $k_n^{-0.373}$  over the inertial range  $3 \leq n \leq 13$  (Figure 1a), consistent with Kolmogorov scaling ( $-1/3$ ) plus the well-documented intermittency correction of the Sabra model [10].

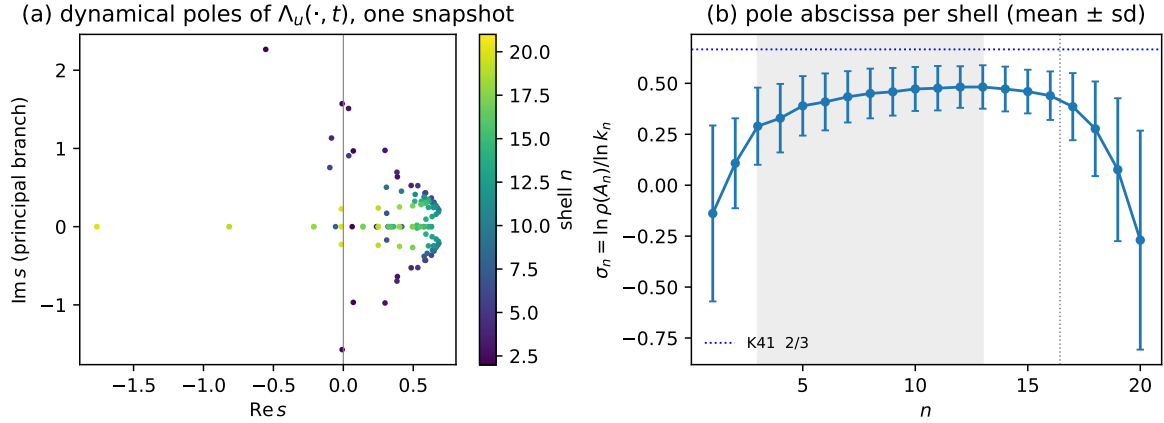


Figure 2: (a) Dynamical poles of  $\Lambda_u(\cdot, t)$  for one snapshot ( $r = 1$ , principal branch), colored by shell. (b) Per-shell pole abscissa  $\sigma_n$  (mean  $\pm$  sd over 1000 snapshots): a plateau across the inertial range (shaded) below the Kolmogorov asymptote  $2/3$  (finite-size offset, Lemma 7.6), then the dissipative plunge.

## 8.2 The pole front measures the cascade exponent

For each snapshot and each admissible  $n$  we form  $A_n^{(r)}$  from Lemma 7.1 (window radii  $r = 1$ :  $6 \times 6$ ;  $r = 2$ :  $10 \times 10$ ) and compute the full spectrum. Figure 2a shows the dynamical poles  $s = \log \alpha / \log k_n$  (principal branch; each pole carries the vertical string (9)) of one snapshot: inertial-range shells cluster along a front at  $\text{Re } s \approx 0.4\text{--}0.6$ , while dissipation-range shells dive to negative abscissae, exactly the drift (10).

Quantitatively, the fitted slope of  $\langle \ln \rho(A_n^{(r)}) \rangle$  against  $\ln k_n$  over the inertial range is

$$\frac{d\langle \ln \rho \rangle}{d \ln k_n} = 0.549 \text{ } (r = 1), \quad 0.542 \text{ } (r = 2),$$

to be compared with the *independently measured* typical-amplitude exponent  $d\langle \ln k_n | u_n | \rangle / d \ln k_n = 0.541$ : agreement within 1.5%, and robust in the window radius. Note that this is the mean-log (typical) exponent, the correct comparison since  $\ln \rho$  is itself a logarithmic quantity; it differs from the rms-based exponent  $1 - 0.373 = 0.627$  and from the non-intermittent Kolmogorov value  $2/3$  precisely by intermittency. The pointwise abscissae  $\sigma_n = \ln \rho(A_n) / \ln k_n$  (Figure 2b) carry, at these moderate  $k_n$ , the finite-size offset  $\log C / \log k_n$  of Lemma 7.6 (here  $C \sim \varepsilon_{\text{typ}}^{1/3} < 1$ ), which is why the front is read off from the slope, not from  $\sigma_n$  itself.

We emphasize the logical order of this section and the next. *Given* the empirical lower bound of §8.3, the tracking of the typical-amplitude exponent by the pole front is an arithmetic consequence of Lemma 7.6, since the matrix entries themselves scale as  $k_n |u_n|$  (Lemma 7.4). The agreement of the two fitted slopes is therefore a consistency check; the nontrivial empirical content is the lower bound itself — the statement that the spectra of these non-normal matrices do not decouple from the amplitudes — to which we now turn.

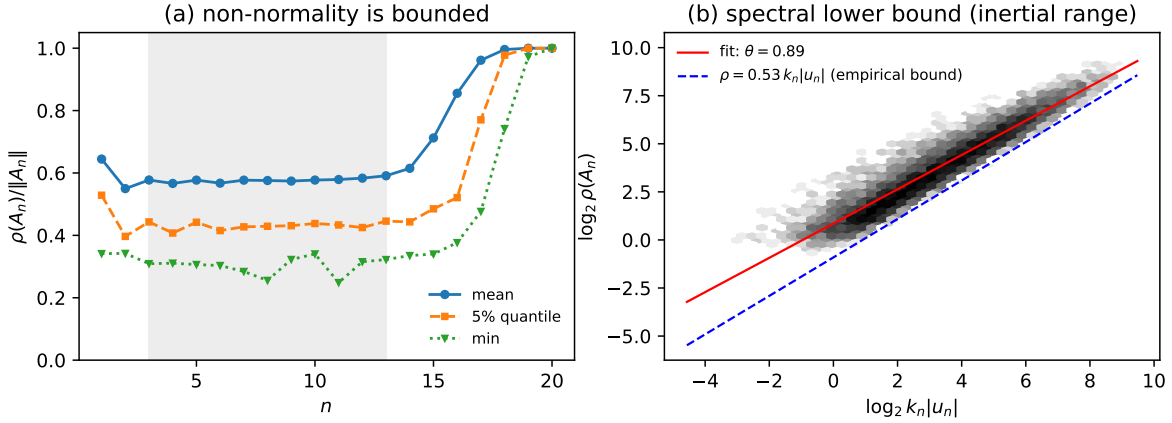


Figure 3: (a) Non-normality index  $\rho(A_n^{(1)})/\|A_n^{(1)}\|$  per shell: mean, 5% quantile, and minimum over 1000 snapshots; the index is bounded away from 0 everywhere. (b) Pooled inertial-range scatter of  $\rho(A_n^{(1)})$  against  $k_n|u_n|$  (log–log density), with the power-law fit ( $\theta = 0.89$ ) and the empirical uniform bound  $\rho \geq 0.53 k_n|u_n|$ .

### 8.3 The lower-bound question in the sandbox

Question 7.7 asks whether the spectrum of the non-normal transfer matrices can “go blind” to the amplitudes (e.g. by near-nilpotency; recall from Remark 6.2 that exact nilpotency would trivialize the determinant entirely). Over all  $11 \times 10^3$  inertial-range samples we find (Figure 3):

**Empirical Finding 8.1.** In the statistically steady state described in §8.1, for all inertial-range shells  $3 \leq n \leq 13$  and all 1000 snapshots, without exception:

$$\frac{\rho(A_n^{(1)})}{\|A_n^{(1)}\|} \geq 0.25 \quad (\text{mean } 0.58, \text{ 5\% quantile } 0.43), \quad \frac{\rho(A_n^{(1)})}{k_n|u_n|} \geq 0.53 \quad (\text{5\% quantile } 0.77),$$

and for  $r = 2$  the corresponding minima are 0.30 and 0.61. A pooled power-law fit gives  $\rho(A_n) \propto (k_n|u_n|)^\theta$  with  $\theta = 0.89$  ( $r = 1$ ),  $0.88$  ( $r = 2$ ).

Thus, empirically, the answer to Question 7.7 in the Sabra sandbox is *affirmative with  $\theta = 1$  and  $c \approx \frac{1}{2}$* : non-normality is present but bounded, and the spectral radius tracks the local amplitude  $k_n|u_n|$  within a factor of about two throughout the inertial range. Combined with Lemma 7.6, this is what licenses reading the cascade exponent off the pole front in §8.2, and it means that, in this model, confinement of the poles of  $\Lambda_u$  is not merely implied by but *equivalent to* control of the shell amplitudes.

### 8.4 Caveats

We state the limits of Finding 8.1 plainly. (1) It is a numerical observation at one parameter set, one forcing, and  $10^3$  snapshots; rare events below the observed minima cannot be excluded, though the 5% quantiles sit well above them. (2) It concerns the *windowed* operators, which Remark 7.2 shows to be the only nontrivial local operators available in a shell model. (3) Above all, it does not transfer to (1): there the shell spaces have dimension  $\sim 2^{3j}$ , the transfer operators

contain the full within-shell advection, and non-normality could a priori degrade with dimension. Question 7.7 for Navier–Stokes remains open; the sandbox result shows only that its answer is not *generically* negative for cascade dynamics.

## 9 Limits of the method

### 9.1 Scaling gap

Under the Navier–Stokes scaling  $u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t)$ , critical spaces (those with scale-invariant norms) include  $\dot{B}_{\infty, \infty}^{-1}$ , the largest reasonable one. The hypothesis of Theorem 5.3 lives in  $\dot{B}_{\infty, \infty}^1$ , *two full derivatives* above critical:  $\|u_\lambda\|_{\dot{B}_{\infty, \infty}^1} = \lambda^2 \|u\|_{\dot{B}_{\infty, \infty}^1}$ . Assuming a subcritical a priori bound is, by the standard heuristic and by Proposition 5.4 rigorously, tantamount to assuming regularity. Any framework whose “tempered” hypothesis normalizes transfer eigenvalues to the unit circle is making a subcritical assumption of exactly this strength.

### 9.2 Tao’s barrier

Every estimate used in this paper — the energy inequality, Bernstein, Kato–Ponce, logarithmic interpolation — depends only on the bilinear, energy-conserving, scale-local structure of the nonlinearity  $\mathbf{P}(u \cdot \nabla u)$ . Tao [12] constructed an averaged version of (1) sharing precisely this structure (same energy identity, same harmonic-analysis estimates) whose solutions blow up in finite time. Hence no argument built solely from such structure — in particular, no analysis of  $\Lambda_u$  resting on the bounds of §§3–5 — can prove global regularity.

### 9.3 What a spectral lower bound would and would not give

Finding 8.1 upgrades the status of Question 7.7 from speculative to plausible, and a proof of its Navier–Stokes form would be a real theorem: together with Lemma 3.4 it would make pole confinement of  $\Lambda_u$  *equivalent* to the  $\dot{B}_{\infty, \infty}^1$  bound of Assumption 5.1, hence, by Theorem 5.3/Proposition 5.4, equivalent to regularity:  $\Lambda_u$  would then be a *faithful* spectral diagnostic — blow-up if and only if the pole front escapes. But faithfulness is not a proof mechanism: by the two barriers above, deciding *which* side of the equivalence holds for the true equations is untouched. Indeed Tao’s averaged system, suitably shell-modeled, would presumably exhibit a faithful  $\Lambda$  whose pole front does escape; faithfulness is symmetric between regularity and blow-up. The open problems this paper leaves are therefore: (a) Question 7.7 for (1), now with quantitative sandbox support ( $\theta = 1$ ,  $c \approx \frac{1}{2}$ , non-normality index  $\geq \frac{1}{4}$ ); and (b) the identification of any dynamically generated structure — a genuine analogue of Poisson summation — that could underwrite a functional equation and break the symmetry that barrier arguments exploit. We do not know how to obtain either.

## 10 Conclusion

Euler-type spectral determinants can be attached rigorously to Navier–Stokes flows (§3); they continue meromorphically to all of  $\mathbf{C}$  along strong solutions, for reasons of parabolic smoothing rather than symmetry (§4); their poles are confined under a tempered-spectrum hypothesis whose exact strength we have identified and shown equivalent to regularity itself (§5); and no functional equation can hold in that regime — for the completed function, and, whenever any

transfer operator has a nonzero eigenvalue, for the bare determinant itself, independently of the archimedean normalization (§6) — which is where the analogy with  $L$ -functions ends. In the Sabra shell model the construction is fully explicit (§7) and fully computed (§8): the pole front of  $\Lambda_u$  advances with the typical-amplitude cascade exponent (0.549 vs. 0.541, within 1.5%), and the spectral lower bound that would make  $\Lambda_u$  a faithful diagnostic holds empirically without exception. What the determinant cannot do is decide regularity: the structural barriers of §9 apply to it in full, and the Millennium Problem is untouched by the present work. We offer the object, its exact dictionary, its delimitation, and one sharp open question.

## Data and code availability

The companion code, which reproduces all figures and enforces the verification checks C1–C4 of §8.1, is available from the author on request (and will be deposited in a public repository upon acceptance).

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